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Machine Learning for Shock-Capturing Numerical Schemes

Learning reconstruction decisions for
1D hyperbolic conservation laws

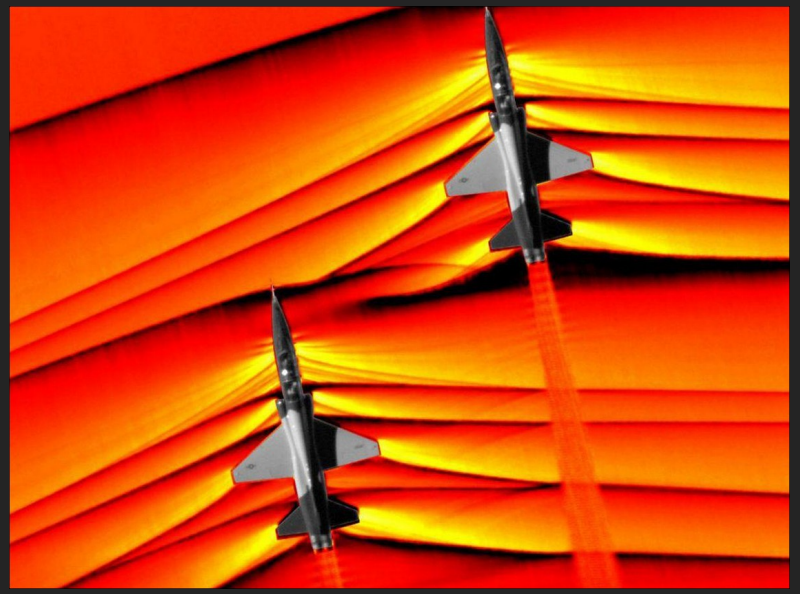
WENO-NN / NN-TENO5 / DIFFERENTIABLE CFD

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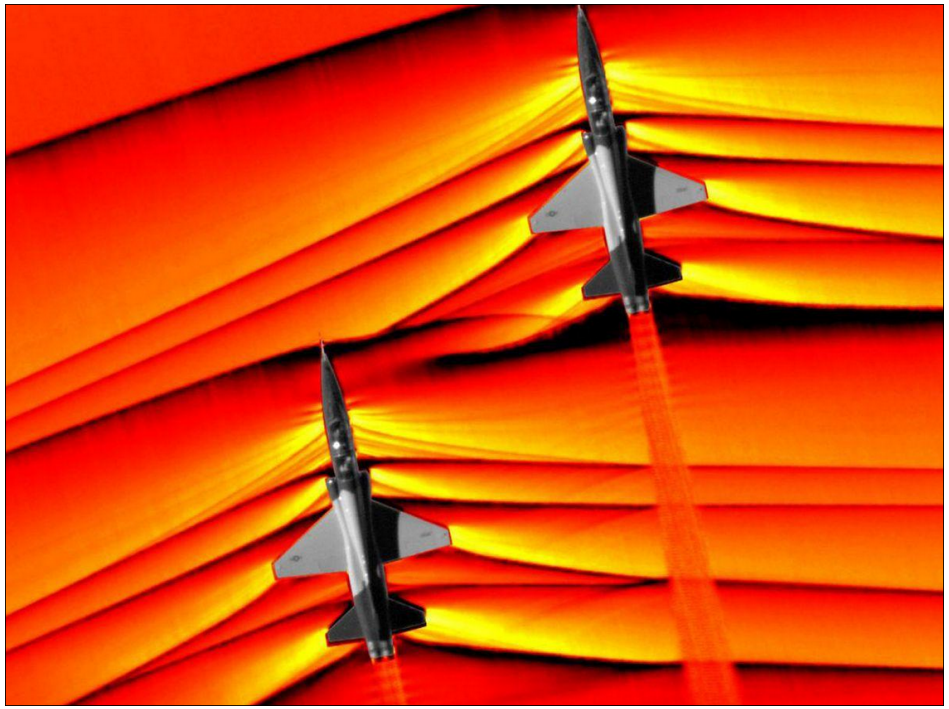
PHYSICS × NUMERICS × LEARNING



Capturing sharp fronts without spurious oscillations

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Learning inside a classical shock-capturing solver



Scientific-ML research internship · UM6P

$$\partial_t U + \partial_x F(U) = 0$$

ENO baseline · WENO/TENO orders 3, 5, 7
Neural reconstruction inside a modular solver
Advection, Burgers, Sod and Shu–Osher

WENO-NN
Coefficient corrections

NN-TENO
Candidate selection

JAX / unrolled
Rollout loss + blending

Later V2
Classical gate + NN

Objective: improve solution accuracy while retaining numerical reliability.

Project scope: src/schemes/ · src/core/ · scripts/legacy_imitation_training/ · scripts/unrolled_physics_loss/

Where the network enters the Euler solver

$$U = (\rho, m, E)^{\mathsf{T}}, \quad F(U) = \begin{pmatrix} m \\ m^2/\rho + p \\ m(E+p)/\rho \end{pmatrix}$$

Conservative forward-Euler building block

$$\bar{U}_i^{n+1} = \bar{U}_i^n - \frac{\Delta t}{\Delta x}\left(\widehat{F}_{i+1/2} - \widehat{F}_{i-1/2}\right)$$

Conserved state U → Roe basis L and R → LF split fluxes → **TENO / NN reconstruction** → Shared-face flux → SSPRK3 stages

Default path: reconstruct characteristic split fluxes, not physical left/right interface states.

$$\mathcal{R}_{V2} = \mathcal{R}_T + G\lambda\left(\mathcal{R}_{NN} - \mathcal{R}_T\right)$$

Later V2
G: classical TENO trouble gate
λ: controlled correction strength

Implemented: euler_jax.py:166–254 · gated_teno5_jax.py:19–67. SSPRK3 reevaluates reconstruction at every stage.

Conservation does not ensure positive Euler states

$$p = (\gamma - 1)\left(E - \frac{m^2}{2\rho}\right) \qquad \rho > 0, \qquad p > 0$$

Oracle NN-TENO5 · Sod, N=120	**Non-positive density · step 6**
V2 soft/.75 · Riemann, N=128	**Negative pressure · step 13, stage 1**
V2 soft/.25 · same short IVP	**15 steps · all raw stages admissible**

Why the attempted safeguards were insufficient

Flux corrections can make internal energy negative.
Normalized weights do not constrain Euler states.
SSPRK3 needs an admissible forward-Euler block.

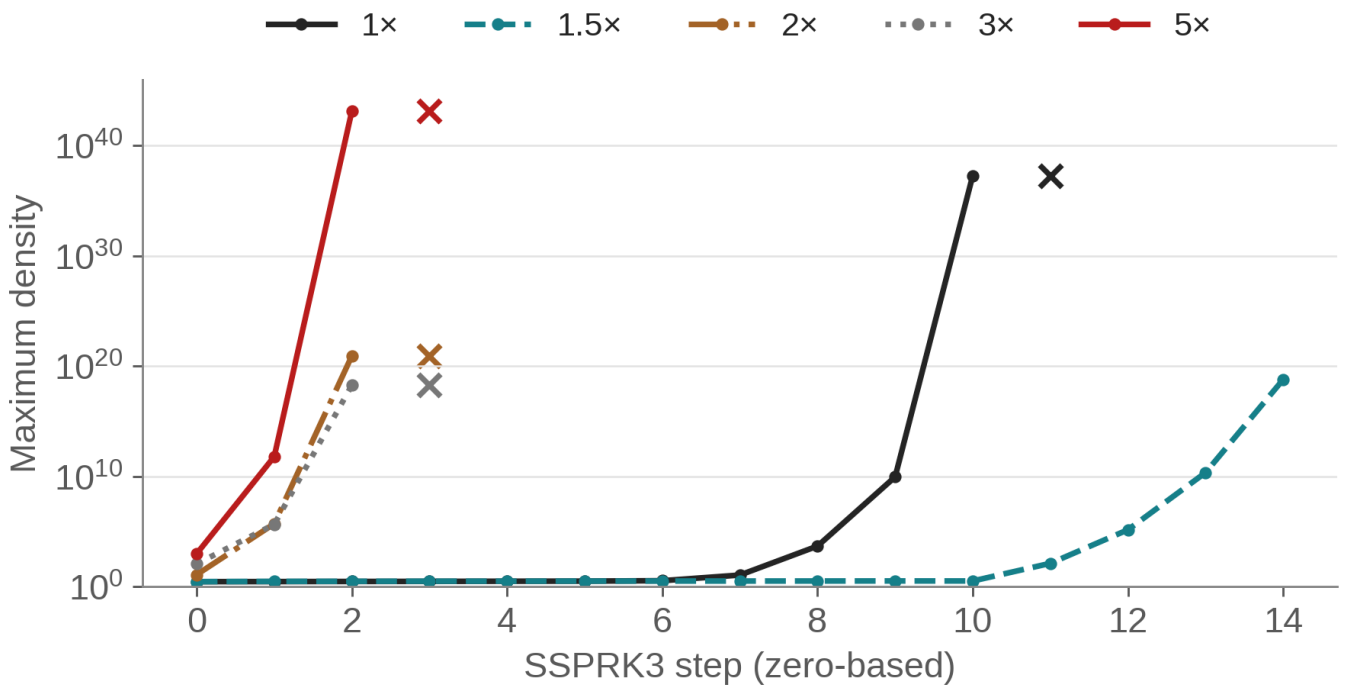
Established remedy · not implemented

Limit shared-face corrections toward an admissible low-order flux, under its CFL.

Observed: audit §§3.1/3.3. Theory: Hu–Adams–Shu (2013); Zhang–Shu (2012); Gottlieb–Shu–Tadmor (2001).

The dissipation sweep did not isolate the cause

Historical Euler training trace · fixed timestep



×: next non-finite step. 1.5× ends finite but enormous.

$$F^{\pm}(U) = \frac{1}{2}\left(F(U) \pm \alpha U\right)$$

$$\mathrm{CFL} \sim \frac{\alpha \Delta t}{\Delta x}$$

Timestep restriction changed

Increasing α at fixed Δt changes stability.

Neural inputs changed

The split-flux stencil also changes.

Roe conditioning was a hypothesis

No condition-number trace isolated it.

Required control: matched CFL, first-invalid-stage diagnostics and cond(R).

Data: models/euler_instability_trace.csv. Test: EulerInstability_TraceCapture.py:66–114. No unique cause established.

Direct state repair breaks the conservative budget

$$\bar{U}_i^{n+1} = \bar{U}_i^n - \frac{\Delta t}{\Delta x}\left(\widehat{F}_{i+1/2} - \widehat{F}_{i-1/2}\right)$$

$$\Delta M_{\text{repair}} = \Delta x \sum_i \left(\rho_i^{\text{after}} - \rho_i^{\text{before}}\right)$$

Implemented · post-update repair

Flux update → Invalid stage → Overwrite cell

RK-weighted repair: Δmass ≈ +0.001611
Δenergy ≈ +0.000889

Post-repair penalties once hid violations.
Guarded training ≠ unguarded validation.

Established · conservative limiting

NN flux → Limit face flux → Update cells

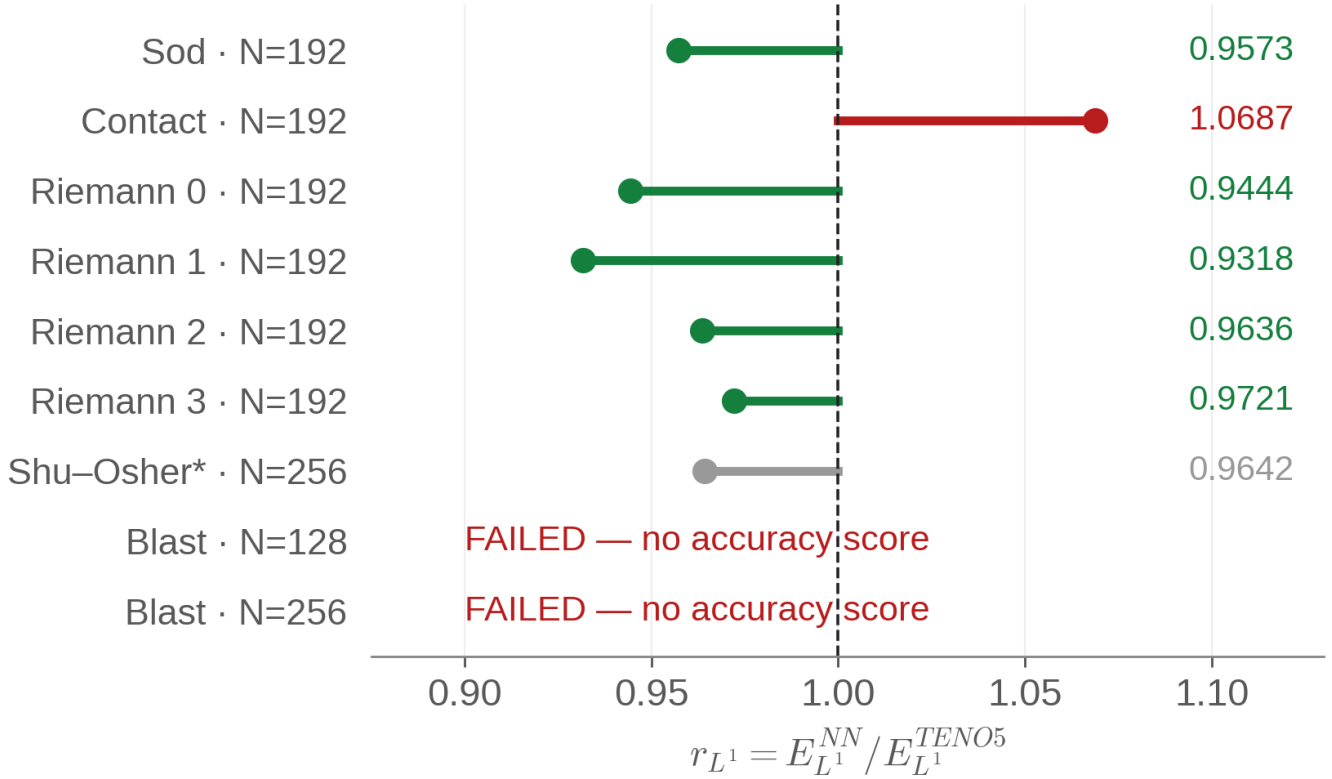
One accepted flux is shared by both cells.
Requires an admissible baseline and stage CFL.

The alternative has not yet been validated
in this neural implementation.

> A finite repaired trajectory does not certify the unguarded learned solver.

Measured: results/repair/repair.json, soft/.75, one repaired stage row. Conservative remedy: Hu–Adams–Shu (2013).

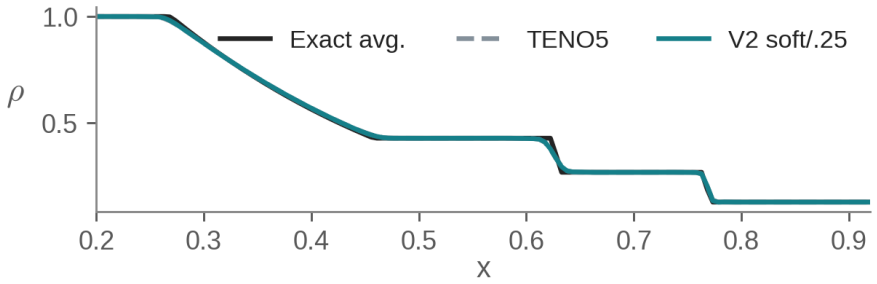
Validation gains did not imply universal transfer



* Shu–Osher ranking: reference uncertainty unresolved.
Blast: no accuracy score; selected model fails at both grids.

0.94794

Validation macro L^1 ratio
307 short Euler IVPs · no invalid stages

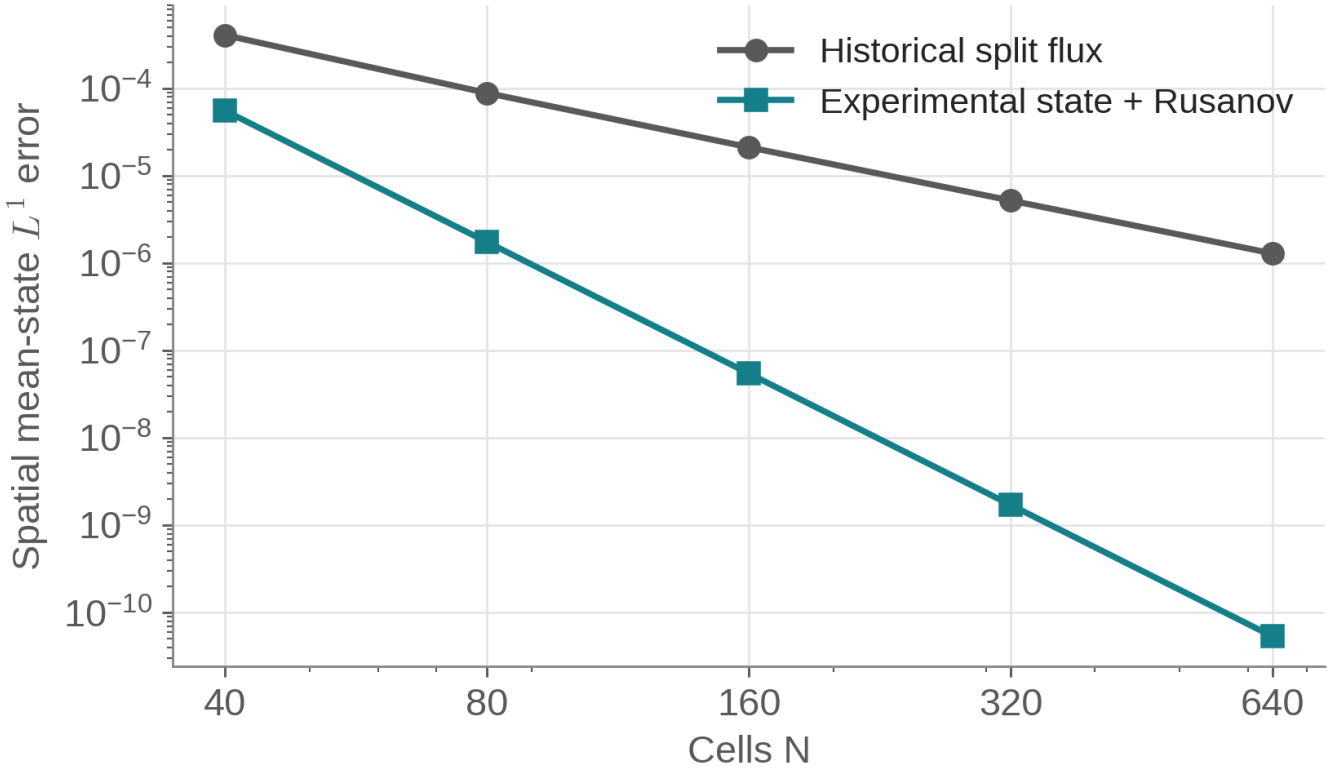


Sod N=192: 4.3% lower L^1, but TV rises and density undershoot ≈ 3.51×10^{-4}.

Candidate shifts: horizon, shock strength, walls.
Warm-up states ≠ deployed split-flux inputs.

Frozen V2 only. Data: full-validation/summary.json · independent/benchmark.json · wall-closure/benchmark.json.

The classical FV pipeline also limited accuracy



$$\overline{F(U)} \neq F(\bar{U})$$

Nonlinear flux evaluation and cell averaging do not commute.

Historical split flux: order ≈ 2.01
State + Rusanov: order ≈ 5.00

Experimental remedy
Consistent FV state reconstruction.
Shock/positivity behavior unvalidated.

> This limitation affects classical TENO5 too; it is not a neural-only failure.

Data: nonlinear_euler.json; semidiscrete exact-average divergence. WENO7 details → Backup B. Wall correction ≠ blast robustness.

Learn within verifiable numerical constraints

Observed limitations

Negative stages · repair sources
Transfer gaps · accuracy/robustness

Established foundations

Positivity · convex limiting
Moments · conditional SSP

Open research

Constrained learned corrections
Independent residual assessment

Shared-face limited flux

$$\widehat{F}^{limited} = \widehat{F}^{LO} + \theta\left(\widehat{F}^{NN} - \widehat{F}^{LO}\right)$$

θ must satisfy stage constraints; an arbitrary value in [0,1] is insufficient.

Independent space–time residual

$$\mathscr{R} = \partial_t \widehat{U} + \partial_x F(\widehat{U})$$

Giesselmann, Kwon & Krumscheid (2026) · RKDG + SIAC

Entropy-compatible convection–diffusion; regular reconstruction and stability analysis required.
This does not certify the present neural Euler solver.

> Can learned high-order reconstruction improve numerical accuracy while retaining provable structural constraints and independently verifiable error control?

Hu–Adams–Shu (2013) · Guermond et al. (2018) · Gottlieb et al. (2001) · arXiv:2604.01200. Discussion; backups follow.

Euler reconstruction and SSPRK3: the full map

Face characteristic decomposition

$$\lambda_j = v - c,\ v,\ v + c, \qquad c = \sqrt{\gamma p/\rho}$$

$$g^{\pm}_{i+1/2,k} = \tfrac{1}{2} L_{i+1/2} \left(F(U_k) \pm a_{i+1/2} U_k \right)$$

$$\widehat{F}_{i+1/2} = R_{i+1/2} \left[\mathcal{R}(g^+) + \mathcal{R}(g^-) \right]$$

Same face basis for both LF directions.
Negative branch: shift by one cell and reverse.
Default a: maximum speed of adjacent cells.

SSPRK3 · admissible at every stage

$$U^{(1)} = U^n - \Delta t D(U^n)$$

$$U^{(2)} = \tfrac{3}{4} U^n + \tfrac{1}{4} \left[U^{(1)} - \Delta t D(U^{(1)}) \right]$$

$$U^{n+1} = \tfrac{1}{3} U^n + \tfrac{2}{3} \left[U^{(2)} - \Delta t D(U^{(2)}) \right]$$

$$\rho > 0, \qquad E - \frac{m^2}{2\rho} > 0$$

D(U) = (F̂right − F̂left)/Δx. Implementation: euler_jax.py:166–254; numerics_jax.py:92–115. SSP: Gottlieb et al. (2001).

WENO7: a defect and an empirical mitigation

$$\alpha_k = \frac{d_k}{(\epsilon + \beta_k)^2}, \qquad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}$$

Reproduced coefficient-layer test

100,000 stored normalized Euler stencils
ε = 10^{-6}: 10 non-finite coefficient rows
ε = 10^{-4}: 0 in the same forward test

Unresolved NaN mechanism

$$(10^{-6})^2 = 10^{-12} \gg 10^{-38}$$

Expanded β polynomials can cancel.
An instrumented near-flat row has β < 0.
Exact NaN operation remains unisolated.

> Epsilon mitigation removed sampled NaNs; it did not prove order or Euler stability.

WenoNetworks.py:106–137 · commit 753d526 · audit §§3.6/10.4. Backend/graph changes can affect bad-row counts.

DG and SIAC: reconstruction before certification

Moment-preserving approximation

$$U_h|_{I_i} = \sum_{\ell=0}^{q} U_{i,\ell}\varphi_\ell(x)$$

$$\widehat{U} = K_h * U_h, \qquad \int K = 1, \qquad \int x^r K(x)\,dx = 0$$

r = 1,…,s: chosen reproduction moments.
Unit mass ≠ cellwise mean preservation.
Signed kernels need not preserve positivity.

Independent residual assessment

$$\mathscr{R} = \partial_t \widehat{U} + \partial_x F(\widehat{U})$$

Relative entropy needs regular reconstruction and suitable admissible state ranges.
Sharp gradients can weaken error bounds.
A residual sensor alone is not a certificate.

2026 paper: periodic Cartesian RKDG + SIAC; convection–diffusion with compatible entropy.
No learned filter or neural Euler shock theorem.

Primary references · full metadata and links in Sources_and_Evidence.md

Guermond et al., SISC 40 (2018) A3211–A3239, doi:10.1137/17M1149961 · Giesselmann–Kwon–Krumscheid, arXiv:2604.01200.
Hu et al., JCP 242 (2013) 169–180 · Zhang & Shu, JCP 231 (2012) 2245–2258 · Gottlieb et al., SIAM Rev. 43 (2001) 89–112.